

Applications

We are now at a point where we can begin to incorporate our pre-trade market impact cost estimates into the investment decision process for stock selection, portfolio optimization, alpha capture, and “what-if” analysis.

The model we will use for our analysis is:

$$MI_{bp} = 793 \cdot \text{Size}^{0.57} \cdot \sigma^{0.78} \cdot \text{POV}^{0.52} \quad (11.5)$$

We next illustrate how our pre-trade of pre-trade model can be used to address various portfolio manager needs. This builds on Journal of Trading, “Creating Dynamic Pretrade Models: Beyond the Black Box,” Fall 2011.

Example 1

An investor wishes to determine expected impact cost for RLK for an order of 10% ADV utilizing a POV = 20% strategy. The volatility of RLK is 20%. Cost estimates were provided by all three brokers (Table 11.1) and are shown below in Table 11.3.

The pre-trade of pre-trade (aggregated model) cost estimate is computed from the following:

$$MI_{bp} = 793 \cdot (0.10)^{0.57} \cdot (0.20)^{0.78} \cdot (0.20)^{0.52} = 26.3 \text{ bp}$$

The average cost from the three brokers is 24.5 bp and the estimated cost using our pre-trade of pre-trade model is 26.3 bp. Notice that the estimate from the pre-trade of pre-trades is consistent with the average of the broker models.

The advantage now is that the average estimated cost across brokers can be computed directly from Equation 11.5 without having to access broker models or shift through broker data. The pre-trade of pre-trades is an important tool for those parties wishing to minimize information leakage.

Table 11.3 Estimated Costs

Broker	Est. Cost
I	24.4
II	29.1
III	19.9
Avg:	24.5
I-Star Model:	26.3